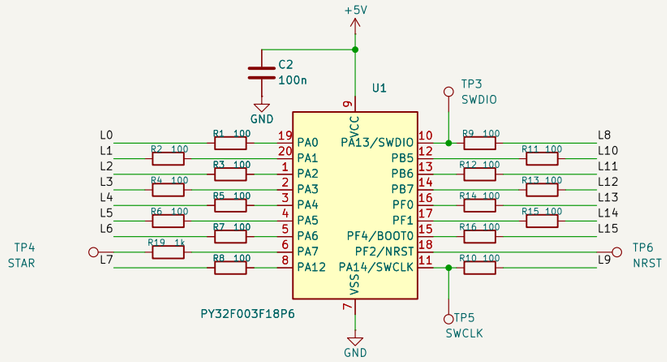
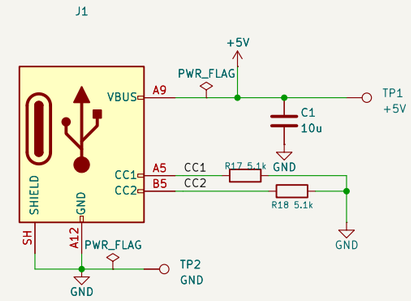
MCU — PY32F003F18P6 + 16 line resistors + touch star



R1-R16: 100 Ω per charlieplex line (keeps the common anode pin < 20 mA).
PA7 = ADC: charge-sharing capacitive touch on the star (hover / touch / press / squeeze).
R19 1 kΩ: ESD protection for the exposed star (EN 61547).
SWD pins PA13/PA14 double as lines L8/L9: program with connect-under-reset (TP3, TP5, TP6).
PF4/BOOT0 = line L15: its reset pull-down keeps boot-from-flash; firmware then floats it.

POWER — USB-C 6-pin, power only, 5 V sink



TYPE-C 6P

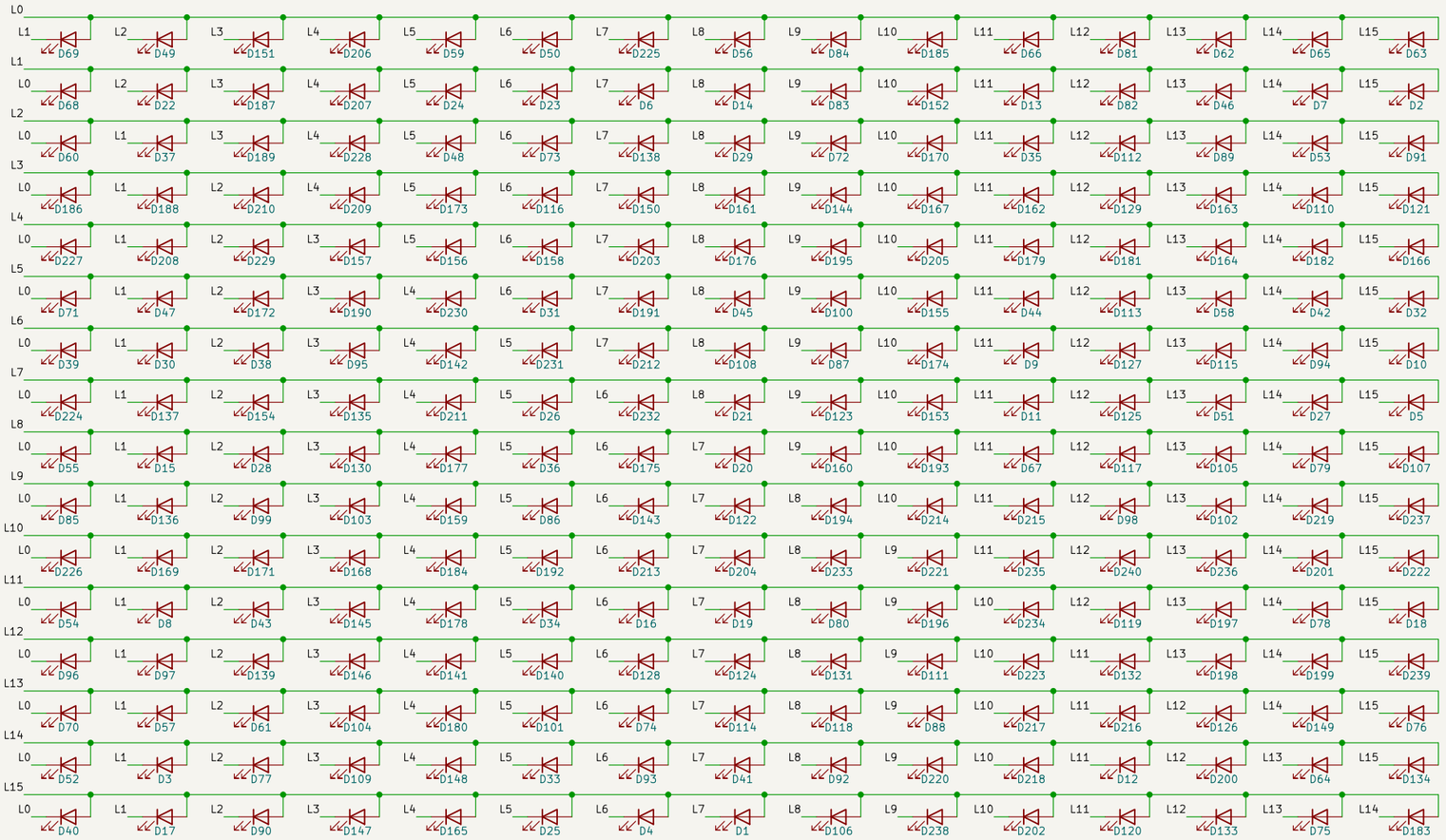
No fuse / TVS / switch: cost-down decision.
Typical draw ≈ 10 mA, max 21 mA @ 5 V.
TP1-TP3, TP5, TP6: bare pads on the back for the pogo-pin SWD jig.

DRIVE SCHEME
Scan by anode: phase p (0..15) drives line Lp HIGH, the cathodes of LEDs lit in that phase LOW,
all other lines Hi-Z. 16 phases per frame, binary-code-modulated PWM (8 bit, 2 µs slot, 122 Hz).
LED Dn+1 sits at board position n (reading order, top->bottom, left->right). Which anode/cathode pair sits at
which position is chosen by scripts/optimize_leds.py (hardware/led_assignment.json): neighbouring LED pads
share line nets, so each net is a few copper islands joined by tiny bridges and the board routes on 2 layers.

TOUCH STAR
The star is one electrode on BOTH copper layers, joined by a plated hole. PA7 samples it with
ADC charge sharing: hover < light touch < firm press < squeeze (pinched front + back = biggest signal).

PRODUCTION
PY32F003F18P6 = TSSOP-20 'pinout 1' (datasheet fig. 3-3). SWD via pogo pads TP3 (SWDIO), TP5 (SWCLK),
TP6 (NRST), connect-under-reset; after boot the firmware takes PA13/PA14 over as lines L8/L9.

LED MATRIX — 240 × XL-1608WWC-06 warm white 0603, charlieplexed on 16 lines



HOW TO READ THE MATRIX

Each row is one charlieplex line driving the ANODES
(bus wire, line name on the left end of the bus).

Each LED's CATHODE goes to the line named
just left of it.

Reference Dn = board position n-1 in reading order
(top -> bottom, left -> right), see the notes above.

L0..L15 connect to R1..R16 in the MCU block.

All LCSC part numbers in the LCSC field of each symbol
Generated by hardware/scripts/gen_sch.py — edit the script, not this file

KTE Project 1

Sheet: /
File: pixelpine.kicad_sch

Title: PixelPine — 240 LED charlieplexed PCB Christmas tree

Size: A3	Date: 2026-09-23	**Rev: A**
KiCad E.D.A. 10.0.0		Id: 1/1